

INDIVIDUAL • INDUSTRY-ORIENTED

CUSTOMER JOURNEY ANALYTICS DASHBOARD

End-to-End Customer Behaviour, Conversion & Revenue Analytics

PROJECT TYPE

Individual End-to-End Data Analytics Project

AUTHOR

Krishan Kumar Sharma

TECHNOLOGY

Excel • Python / Jupyter • pandas • MySQL • SQL • Power BI

ANALYTICAL DATASET

5,982 records • 27 columns

2026

Project at a Glance

Item	Project Detail
Project	Customer Journey Analytics Dashboard
Owner	Krishan Kumar Sharma
Project Type	Individual, end-to-end industry-oriented analytics project
Primary Objective	Convert customer journey data into validated analytical outputs and an interactive decision-support dashboard
Analytical Dataset	5,982 records × 27 columns
Analytical Database	MySQL / customer_journey table
Visualization Platform	Power BI
Dashboard Pages	5
Primary Domains	Customer behaviour, purchase conversion, traffic performance and revenue analysis

Executive Summary

Customer Journey Analytics Dashboard is an individual end-to-end analytics project developed by Krishan Kumar Sharma to transform customer journey data into a practical business intelligence solution. The workflow begins with Excel-based data understanding, moves through profiling, cleaning, standardization and validation in Python/Jupyter, continues with MySQL and SQL analysis, and ends with a five-page Power BI dashboard.

For this revised submission, the analytical dataset size is explicitly corrected to 5,982 records. These are the records that were loaded into Power BI, and the SQL/Power BI computations discussed in this report are based on that same analytical population.

The solution emphasizes traceability: data-quality work precedes business interpretation, SQL provides reproducible analytical outputs, and Power BI communicates the findings through focused pages covering overview, customer behaviour, purchase analysis, traffic and revenue, and key insights.

Project Outcome

- 5,982-record analytical dataset used for SQL and Power BI analysis.
- 27 analytical columns supporting customer, session, purchase, traffic, product and revenue analysis.
- Data-quality checks covering missing values, categorical consistency and duplicate records.
- MySQL-based SQL analysis covering purchase, conversion, traffic, category, revenue and payment behaviour.
- Five-page Power BI dashboard for interactive exploration and decision support.
- Business insights and recommendations connecting customer behaviour, acquisition efficiency and revenue.

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1. Project Background and Business Context

Digital customer journeys generate large amounts of behavioural and transaction-oriented information. A session may originate from a traffic source, involve different devices and engagement levels, progress toward a product interaction or purchase, and generate revenue only for a subset of sessions. Analysing these elements independently can hide the relationship between acquisition, engagement, conversion and commercial value.

This project was developed as an individual, industry-oriented analytics solution to connect these dimensions into one traceable workflow. The objective is to turn customer journey data into an analytical asset, answer practical business questions using SQL, and communicate the results through an interactive Power BI dashboard.

2. Problem Definition

The raw dataset required systematic examination before it could be used for reporting. The documented preparation work identified categorical inconsistencies, duplicate records, invalid rating values and missing values. If such issues are carried directly into analysis, one business category can be split into multiple labels, duplicates can inflate counts, and structurally unavailable fields can be misinterpreted as data-entry failures.

The project therefore addresses four connected problems: understanding the source data, improving its quality without losing meaningful context, extracting reliable metrics, and presenting those metrics in a form that supports interpretation and decision-making.

3. Objectives

- Understand the structure, fields and analytical meaning of the customer journey dataset.
- Profile the dataset before transformation to identify quality issues.
- Standardize inconsistent categorical values and validate important analytical fields.
- Identify and remove duplicate records while preserving the intended analytical structure.
- Distinguish ordinary missing data from structurally non-applicable values where business context supports that interpretation.
- Validate the analytical dataset used for database and dashboard analysis.
- Use MySQL and SQL to answer customer, conversion, traffic, category, revenue and payment questions.
- Develop a five-page Power BI dashboard that communicates the customer journey from overview to detailed analysis and key insights.

4. Scope and Deliverables

- Raw Excel dataset and data-dictionary review.
- Python/Jupyter-based profiling and cleaning evidence.
- Analytical dataset used for downstream SQL and Power BI analysis.
- MySQL database import and business-focused SQL analysis.
- Five-page Power BI dashboard.
- Integrated business insights, recommendations, limitations and future scope.
- Professional GitHub project documentation.

5. End-to-End Solution Architecture

ANALYTICS FLOW

RAW EXCEL DATA → DATA UNDERSTANDING → DATA PROFILING → CLEANING / STANDARDIZATION → ANALYTICAL VALIDATION → MYSQL IMPORT → SQL BUSINESS ANALYSIS → POWER BI DASHBOARD → INSIGHTS & RECOMMENDATIONS

The architecture is deliberately sequential. Data-quality work precedes SQL aggregation, and SQL findings are then communicated through Power BI.

6. Dataset and Data Understanding

The source is session-oriented customer journey data. The documented fields cover customer and visit attributes, demographic information, customer type, previous purchases, device, traffic source, engagement measures, product/category information, cart and discount behaviour, purchase status, purchase amount, payment method, delivery type, review rating and customer satisfaction.

The critical analytical boundary for this revised report is explicit: **5,982 records were loaded into Power BI, and all dashboard computations were performed on those 5,982 records.**

Dimension	Examples of analytical use
Customer	New vs returning behaviour, customer counts, previous purchases
Session	Session duration, engagement and conversion context
Acquisition	Traffic source sessions, purchases, conversion and revenue
Product	Category purchase volume and category revenue
Transaction	Purchase amount, payment method and discount usage
Device	Sessions, purchases, revenue and conversion by device
Feedback	Review rating and customer satisfaction where applicable

Session_ID	Customer_ID	Visit_Date	Age	Gender	City	Customer_Type	Previous_Purchases	Device	Traffic_Source	Session_Duration_Min	Pages_Viewed	Products_Viewed	Product_Category	Item_Purchased
SESS05319	CUST11585	31-10-2025	28	Female	Lucknow	New	1	Desktop	Facebook	17.7	6	2	Fashion	T-Shirt
SESS03219	CUST10452	14-02-2026	43	Male	Lucknow	Returning	8	Mobile	Facebook	9.9	5	2	Electronics	Headphone
SESS05118	CUST11821	29-06-2025	36	Male	Jalpur	New	0	Desktop	Facebook	6.7	10	3	Beauty	Shampoo
SESS01667	CUST10036	24-02-2026	26	Male	Delhi	New	0	Desktop	Google	4.5	6	5	Books	Academic B
SESS03660	CUST10940	24-09-2025	36	Female	Hyderabad	New	0	Mobile	Email	11.2	6	4	Home & Kitchen	Table Lamp
SESS05628	CUST11999	28-08-2025	39	Male	Ahmedabad	Returning	11	Tablet	Referral	7.9	1	4	fashion	Sneakers
SESS05138	CUST11985	31-05-2026	18	Male	Pune	New	0	Mobile	Instagram	10.6	8	3	Fashion	T-Shirt
SESS02446	CUST10579	03-10-2025	37	Female	Hyderabad	Returning	13	Mobile	YouTube	6	8	4	Fashion	Sneakers
SESS05253	CUST11739	05-02-2026	25	Male	Delhi	New	0	Mobile	Direct	11.6	5	7	Beauty	Face Wash
SESS01371	CUST10729	19-03-2026	40	Female	Delhi	New	1	Desktop	Direct	3.8	7	3	Books	Academic B
SESS02143	CUST11206	10-09-2025	37	Male	Ahmedabad	Returning	10	Mobile	Direct	4.1	8	3	Beauty	Shampoo
SESS05681	CUST10156	12-01-2026	26	Male	Chennai	Returning	11	Mobile	Direct	10.2	6	8	Home & Kitchen	Storage Box
SESS00692	CUST11255	10-07-2025	31	Male	Bangalore	New	0	Desktop	Instagram	5	7	4	Electronics	Headphone
SESS01965	CUST10652	17-06-2026	42	Female	Delhi	New	0	Mobile	Google	8	6	6	Fashion	T-Shirt
SESS03616	CUST11012	24-06-2025	18	Male	Mumbai	New	0	Mobile	Referral	2.2	7	3	Home & Kitchen	Storage Box
SESS00697	CUST12073	07-11-2025	25	Female	Pune	Returning	3	Mobile	YouTube	5.9	4	5	Beauty	Moisturizer
SESS00531	CUST12037	14-01-2026	43	Male	Bangalore	New	0	Tablet	Google	6.9	7	6	Electronics	Power Bank
SESS05101	CUST10426	14-03-2025	20	Male	Delhi	Returning	5	Mobile	Google	3.4	5	3	Home & Kitchen	Bedsheet
SESS03986	CUST11493	16-03-2025	38	Female	Mumbai	Returning	15	Mobile	Instagram	12.3	6	3	Books	Academic B
SESS03327	CUST10061	23-01-2025	38	Female	Kolkata	New	1	Mobile	Direct	3.1	9	2	Sports	Cricket Bat
SESS05431	CUST11587	09-05-2026	28	Male	Ahmedabad	New	1	Mobile	Google	18.8	4	6	Sports	Dumbbells
SESS01919	CUST10789	23-02-2025	35	Male	Pune	New	0	Mobile	Direct	11.3	10	3	Electronics	Power Bank
SESS03746	CUST11084	20-03-2026	32	Female	Ahmedabad	Returning	13	Desktop	Email	12.9	7	2	Home & Kitchen	Bedsheet
SESS00764	CUST12040	03-02-2025	51	Male	Bangalore	New	0	Mobile	Google	12.1	8	6	Home & Kitchen	Coffee Mak
SESS03066	CUST10552	05-02-2026	18	Female	Hyderabad	Returning	4	Mobile	Instagram	14.9	7	3	Electronics	Wireless E

Figure 1. Raw customer journey dataset

Column_Name	Description
Session_ID	Unique identifier for each browsing session
Customer_ID	Unique customer identifier
Visit_Date	Date on which the customer visited
Age	Customer age in years
Gender	Customer gender
City	Customer city
Customer_Type	New or returning customer
Previous_Purchases	Number of purchases made before this session
Device	Device used for browsing
Traffic_Source	Source through which customer arrived
Session_Duration_Min	Length of browsing session in minutes
Pages_Viewed	Number of pages viewed
Products_Viewed	Number of products viewed
Product_Category	Product category
Item_Purchased	Product/Item viewed or purchased
Product_Price_INR	Base product price in INR
Added_to_Cart	Whether a product was added to cart
Cart_Value_INR	Value of items placed in cart
Discount_Offered	Whether a discount was offered
Discount_Percent	Discount percentage offered
Discount_Used	Whether customer used the discount

Figure 2. Dataset data dictionary

```
[2]: import pandas as pd
```

```
[4]: !pip install openpyxl
```

Collecting openpyxl

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip

Downloading openpyxl-3.1.5-py2.py3-none-any.whl.metadata (2.5 kB)
Collecting et_xmlfile (from openpyxl)
Downloading et_xmlfile-2.0.0-py3-none-any.whl.metadata (2.7 kB)
Downloading openpyxl-3.1.5-py2.py3-none-any.whl (250 kB)
Downloading et_xmlfile-2.0.0-py3-none-any.whl (18 kB)
Installing collected packages: et_xmlfile, openpyxl
Successfully installed et_xmlfile-2.0.0 openpyxl-3.1.5

```
[5]: df = pd.read_excel("Inside_the_Customer_Journey_Raw_Dataset.xlsx")
```

```
[6]: df.head()
```

	Session_ID	Customer_ID	Visit_Date	Age	Gender	City	Customer_Type	Previous_Purchases	Device	Traffic_Source	...	Cart_Value_INR	Discount_Offered	Dis
0	SESS05319	CUST111585	2025-10-31	28	Female	Lucknow	New	1	Desktop	Facebook	...	10537	Yes	
1	SESS03219	CUST10452	2026-02-14	43	Male	Lucknow	Returning	8	Mobile	Facebook	...	59330	Yes	
2	SESS05118	CUST111821	2025-06-29	36	Male	Jaipur	New	0	Desktop	Facebook	...	0	No	

Figure 3. Initial dataset inspection

```
[9]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 6035 entries, 0 to 6034

Data columns (total 27 columns):

#	Column	Non-Null Count	Dtype
0	Session_ID	6035 non-null	object
1	Customer_ID	6035 non-null	object
2	Visit_Date	6035 non-null	datetime64[ns]
3	Age	6035 non-null	int64
4	Gender	5990 non-null	object
5	City	6000 non-null	object
6	Customer_Type	6035 non-null	object
7	Previous_Purchases	6035 non-null	int64
8	Device	6035 non-null	object
9	Traffic_Source	6005 non-null	object
10	Session_Duration_Min	6035 non-null	float64
11	Pages_Viewed	6035 non-null	int64
12	Products_Viewed	6035 non-null	int64
13	Product_Category	6035 non-null	object
14	Item_Purchased	6035 non-null	object
15	Product_Price_INR	6035 non-null	float64
16	Added_to_Cart	6035 non-null	object
17	Cart_Value_INR	6035 non-null	int64
18	Discount_Offered	6035 non-null	object
19	Discount_Percent	6035 non-null	int64
20	Discount_Used	6010 non-null	object
21	Purchased	6035 non-null	object
22	Purchase_Amount_INR	6035 non-null	object
23	Payment_Method	6035 non-null	object

Figure 4. Dataset structure and data types

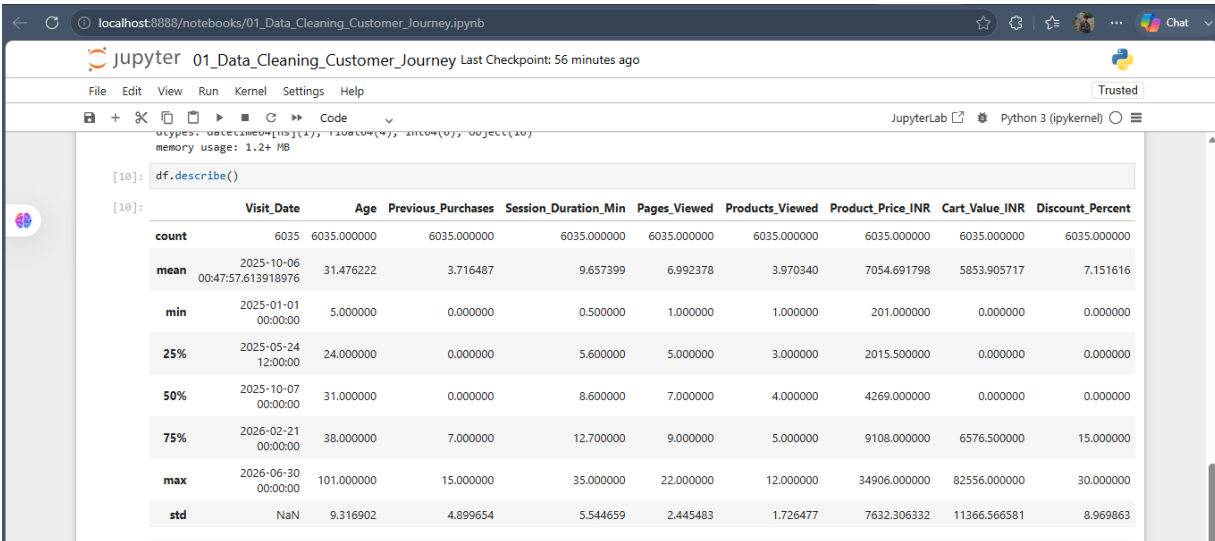


Figure 5. Descriptive statistics

7. Data Profiling

Initial profiling was performed before final analytical use. The evidence covers dataset inspection, data types, descriptive statistics, missing-value counts, unique values and frequency distributions for categorical variables. The purpose was to identify issues that could distort downstream SQL and dashboard metrics.

7.1 Missing-Value Analysis

Missingness was assessed at column level. Review_Rating and Customer_Satisfaction show substantial missingness. These fields require business-context interpretation because a customer who did not purchase would not normally be expected to provide a review or satisfaction response. The project therefore treats much of this missingness as structurally not applicable rather than simply deleting or arbitrarily imputing the values.

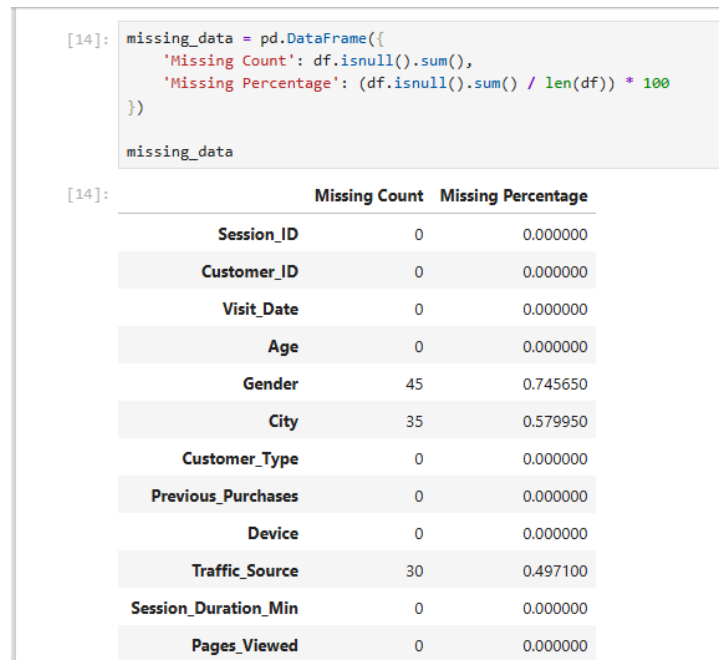


Figure 6. Missing-value analysis

7.2 Categorical Consistency

Frequency inspection identified inconsistent representations in Product_Category, Gender and City. Standardization was applied to reduce split categories and improve the reliability of grouped SQL and Power BI analysis.

```
[28]: df['Product_Category'].value_counts(dropna=False)
```

```
[28]: Product_Category
Fashion          1435
Electronics      1228
Beauty           941
Home & Kitchen   881
Books            715
Sports           653
fashion          47
electronics       39
home & kitchen    26
sports           25
beauty           24
books            21
Name: count, dtype: int64
```

Figure 7. Product category before standardization

```
[29]: df['Product_Category'] = df['Product_Category'].str.strip().str.title()
```

```
[30]: df['Product_Category'].value_counts()
```

```
[30]: Product_Category
Fashion          1482
Electronics      1267
Beauty           965
Home & Kitchen   907
Books            736
Sports           678
Name: count, dtype: int64
```

Figure 8. Product category after standardization

```
[33]: df['Gender'].value_counts(dropna=False)
```

```
[33]: Gender
Male          3253
Female        2523
female         45
NaN            45
FEMALE         44
F              35
male           35
M              32
MALE           23
Name: count, dtype: int64
```

Figure 9. Gender before standardization

```
[37]: df['Gender'].value_counts(dropna=False)
```

```
[37]: Gender
      MALE      3343
      FEMALE    2647
      NaN         45
      Name: count, dtype: int64
```

Figure 10. Gender after standardization

```
[40]: City
      Delhi      1029
      Mumbai     881
      Bangalore  764
      Hyderabad  523
      Pune       506
      Chennai    498
      Kolkata    454
      Jaipur     425
      Ahmedabad  392
      Lucknow    286
      delhi      48
      Delhi      37
      NaN        35
      MUMBAI     35
      DELHI      32
      bangalore  32
      New Delhi  32
      mumbai     26
      Name: count, dtype: int64
```

Figure 11. City before standardization

```
[42]: City
      Delhi      1146
      Mumbai     942
      Bangalore  796
      Hyderabad  523
      Pune       506
      Chennai    498
      Kolkata    454
      Jaipur     425
      Ahmedabad  392
      Lucknow    286
      NaN        35
      New Delhi  32
      Name: count, dtype: int64
```

Figure 12. City after standardization

8. Data Cleaning and Transformation

8.1 Product Category Standardization

Product_Category contained inconsistent capitalization and whitespace. The cleaning approach removed leading/trailing spaces and standardized category labels into a consistent presentation format. This prevents one business category from appearing as multiple groups during aggregation.

8.2 Gender and City Standardization

Gender and City were reviewed through before/after frequency outputs. The objective was to consolidate inconsistent textual representations and make downstream grouping more reliable.

8.3 Review Rating Validation

Invalid Review_Rating values identified during profiling were converted to missing values rather than replaced with arbitrary scores. Remaining missing values were then interpreted using purchase/non-purchase context.

8.4 Duplicate Handling

Duplicate records were explicitly checked and removed during data preparation. For this revised submission, the downstream analytical boundary is the 5,982-record dataset that was actually loaded into Power BI and used for computation.

```
[62]: Review_Rating
```

0.0	2
1.5	2
1.6	1
1.8	1
2.0	3
2.1	3
2.2	9
2.3	6
2.4	8
2.5	18
2.6	15
2.7	17
2.8	34
2.9	46
3.0	42
3.1	67
3.2	59
3.3	76
3.4	98
3.5	92
3.6	110
3.7	117
3.8	124
3.9	147
4.0	160
4.1	123
4.2	113
4.3	119
4.4	131
4.5	106

Figure 13. Duplicate inspection during cleaning

8.5 Structural Missing Values

Review_Rating and Customer_Satisfaction nulls are largely associated with non-purchasing sessions. These values are therefore documented as structurally non-applicable in the analytical narrative rather than being described as ordinary errors.

CONTROL NOTE. Per the project execution, 5,982 records were loaded into Power BI and all downstream computations were performed on those records.

9. Final Analytical Validation

The final analytical boundary for this submission is the dataset loaded into Power BI: **5,982 records × 27 columns**. The SQL purchase-status output independently reconciles to this session count because 2,274 purchased sessions plus 3,708 non-purchased sessions equals 5,982.

Validation Item	Revised Analytical Result
Records used in Power BI	5,982
Columns	27
Purchased sessions	2,274
Non-purchased sessions	3,708
Overall conversion	38.01%
Total SQL revenue	₹23,454,694
Average purchase value	₹10,314.29

10. MySQL Database and SQL Analysis

The analytical dataset was imported into MySQL Workbench for structured SQL analysis. The database layer provides a reproducible environment for answering business questions around purchase status, customer type, traffic source, product category, revenue and payment methods.

```
[117]: print("FINAL DATASET VALIDATION")
print("-----")
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
print("Duplicate Rows:", df.duplicated().sum())
print("Total Missing Values:", df.isnull().sum().sum())
print("Age Range:", df['Age'].min(), "to", df['Age'].max())
print("Purchase Amount Min:", df['Purchase_Amount_INR'].min())

FINAL DATASET VALIDATION
-----
Rows: 6000
Columns: 27
Duplicate Rows: 0
Total Missing Values: 7453
Age Range: 18.0 to 65.0
Purchase Amount Min: 0.0
```

Figure 14. MySQL database setup

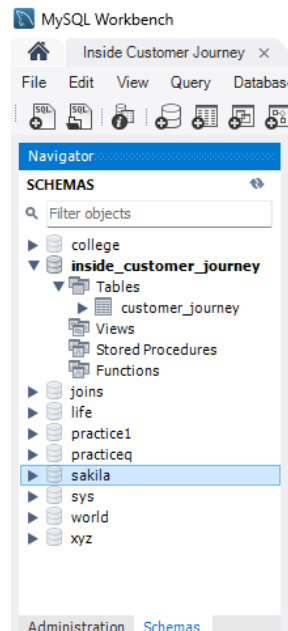


Figure 15. SQL — Purchased vs Not Purchased

```

31 • SELECT
32     Purchased,
33     COUNT(*) AS total_sessions
34 FROM customer_journey
35 GROUP BY Purchased;

```

Result Grid | Filter Rows:

	Purchased	total_sessions
▶	Yes	2274
	No	3708

Figure 16. SQL — Conversion Rate by Customer Type

Query 1

Limit to 1000 rows

```

36 • SELECT
37     Customer_Type,
38     COUNT(*) AS total_sessions,
39     SUM(CASE WHEN Purchased = 'Yes' THEN 1 ELSE 0 END) AS purchases,
40     ROUND(
41         SUM(CASE WHEN Purchased = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
42         2
43     ) AS conversion_rate
44 FROM customer_journey
45 GROUP BY Customer_Type;

```

Result Grid | Filter Rows: | Export: | Wrap C

	Customer_Type	total_sessions	purchases	conversion_rate
▶	New	3405	1150	33.77
	Returning	2577	1124	43.62

Figure 17. SQL — Traffic Source Conversion

Query 1

```

48 COUNT(*) AS total_sessions,
49 SUM(CASE WHEN Purchased = 'Yes' THEN 1 ELSE 0 END) AS purchases,
50 ROUND(
51     SUM(CASE WHEN Purchased = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
52     2
53 ) AS conversion_rate
54 FROM customer_journey
55 GROUP BY Traffic_Source
56 ORDER BY conversion_rate DESC;

```

Result Grid

Traffic_Source	total_sessions	purchases	conversion_rate
Facebook	663	268	40.42
Google	1501	576	38.37
Email	616	235	38.15
Direct	985	374	37.97
Instagram	1067	403	37.77
YouTube	607	223	36.74
Referral	513	188	36.65
Unknown	30	7	23.33

Figure 18. SQL — Product Category Performance

Query 1

```

57 SELECT
58     Product_Category,
59     COUNT(*) AS total_sessions,
60     SUM(CASE WHEN Purchased = 'Yes' THEN 1 ELSE 0 END) AS purchases,
61     ROUND(
62         SUM(CASE WHEN Purchased = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
63         2
64     ) AS conversion_rate
65 FROM customer_journey
66 GROUP BY Product_Category
67 ORDER BY purchases DESC;

```

Result Grid

Product_Category	total_sessions	purchases	conversion_rate
Fashion	1472	572	38.86
Electronics	1256	478	38.06
Beauty	961	374	38.92
Home & Kitchen	901	326	36.18
Books	724	276	38.12
Sports	668	248	37.13

Figure 19. SQL — Revenue by Product Category

Query 1

```

57 SELECT
58     Product_Category,
59     COUNT(*) AS total_sessions,
60     SUM(CASE WHEN Purchased = 'Yes' THEN 1 ELSE 0 END) AS purchases,
61     ROUND(
62         SUM(CASE WHEN Purchased = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
63         2
64     ) AS conversion_rate
65 FROM customer_journey
66 GROUP BY Product_Category
67 ORDER BY purchases DESC;
68 -- SQL-07: Revenue by Product Category
69
70 SELECT

```

Result Grid

Product_Category	total_sessions	purchases	total_revenue	avg_purchase_value
Electronics	1256	478	12366407	25871.14
Home & Kitchen	901	326	3548357	10884.53
Fashion	1472	572	3038050	5311.28
Sports	668	248	2424640	9776.77
Beauty	961	374	1515405	4051.89
Books	724	276	561835	2035.63

Figure 20. SQL — Revenue by Traffic Source

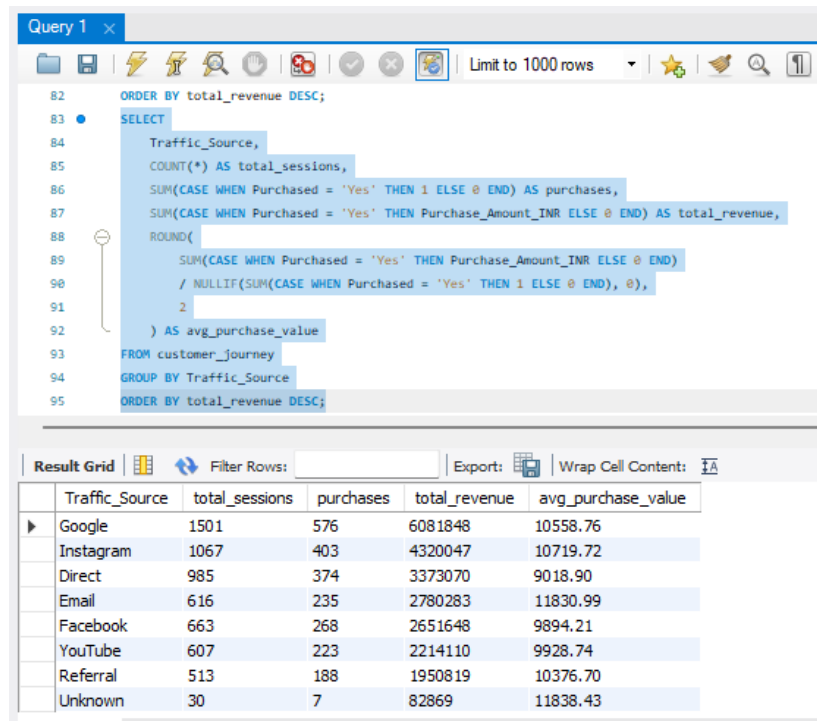


Figure 21. SQL — Successful Purchases by Payment Method

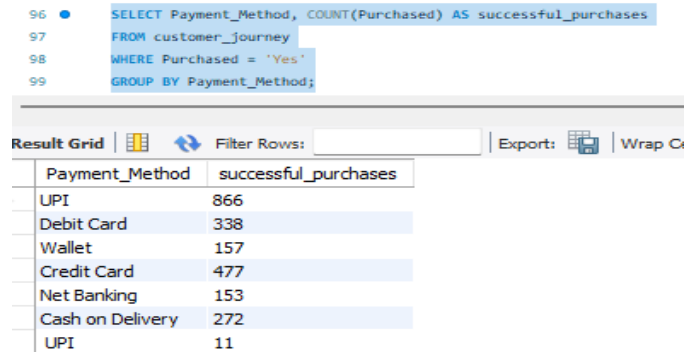


Figure 22. SQL — Overall Purchase and Revenue Performance

10.1 SQL Findings

Analysis Area	Verified Result
Purchased vs Not Purchased	2,274 purchased sessions; 3,708 non-purchased sessions.
Customer Type Conversion	New: 33.77%; Returning: 43.62%.
Traffic Source Conversion	Facebook leads at 40.42%; Google follows at 38.37%.
Product Category Performance	Fashion leads purchase count with 572 purchases.
Revenue by Product Category	Electronics leads with ₹12,366,407.
Revenue by Traffic Source	Google leads with ₹6,081,848.
Payment Method	Credit Card 477; Debit Card 338; COD 272; Wallet 157; Net Banking 153; UPI-labelled rows require review.
Overall Performance	2,274 purchases; ₹23,454,694 revenue; ₹10,314.29 average purchase value.

10.2 Customer Type Conversion

Customer Type	Sessions	Purchases	Conversion
New	3,405	1,150	33.77%
Returning	2,577	1,124	43.62%

10.3 Traffic Source Conversion

Traffic Source	Sessions	Purchases	Conversion
Facebook	663	268	40.42%
Google	1,501	576	38.37%
Email	616	235	38.15%
Direct	985	374	37.97%
Instagram	1,067	403	37.77%
YouTube	607	223	36.74%
Referral	513	188	36.65%
Unknown	30	7	23.33%

10.4 Product Category Performance

Category	Sessions	Purchases	Conversion
Fashion	1,472	572	38.86%
Electronics	1,256	478	38.06%
Beauty	961	374	38.92%
Home & Kitchen	901	326	36.18%
Books	724	276	38.12%
Sports	668	248	37.13%

10.5 Revenue by Product Category

Category	Revenue	Avg. Purchase Value
Electronics	₹12,366,407	₹25,871.14
Home & Kitchen	₹3,548,357	₹10,884.53
Fashion	₹3,038,050	₹5,311.28
Sports	₹2,424,640	₹9,776.77
Beauty	₹1,515,405	₹4,051.89
Books	₹561,835	₹2,035.63

10.6 Revenue by Traffic Source

Traffic Source	Revenue	Avg. Purchase Value
Google	₹6,081,848	₹10,558.76
Instagram	₹4,320,047	₹10,719.72
Direct	₹3,373,070	₹9,018.90
Email	₹2,780,283	₹11,830.99
Facebook	₹2,651,648	₹9,894.21
YouTube	₹2,214,110	₹9,928.74
Referral	₹1,950,819	₹10,376.70
Unknown	₹82,869	₹11,838.43

11. Power BI Dashboard

The final analytical layer is a five-page Power BI dashboard. The dashboard acts as the communication layer: SQL provides precise analytical outputs, while Power BI organizes the information into interactive views for exploration and decision support.

ANALYTICAL DATA BOUNDARY

The Power BI dashboard was loaded with 5,982 records. All dashboard calculations and visual outputs discussed in this report were computed on that 5,982-record analytical dataset.

Separate Power Query, DAX-measure and relationship-model screenshots are not included in the supplied project evidence. Accordingly, this report documents the dashboard pages and their analytical purpose without inventing undocumented implementation details.

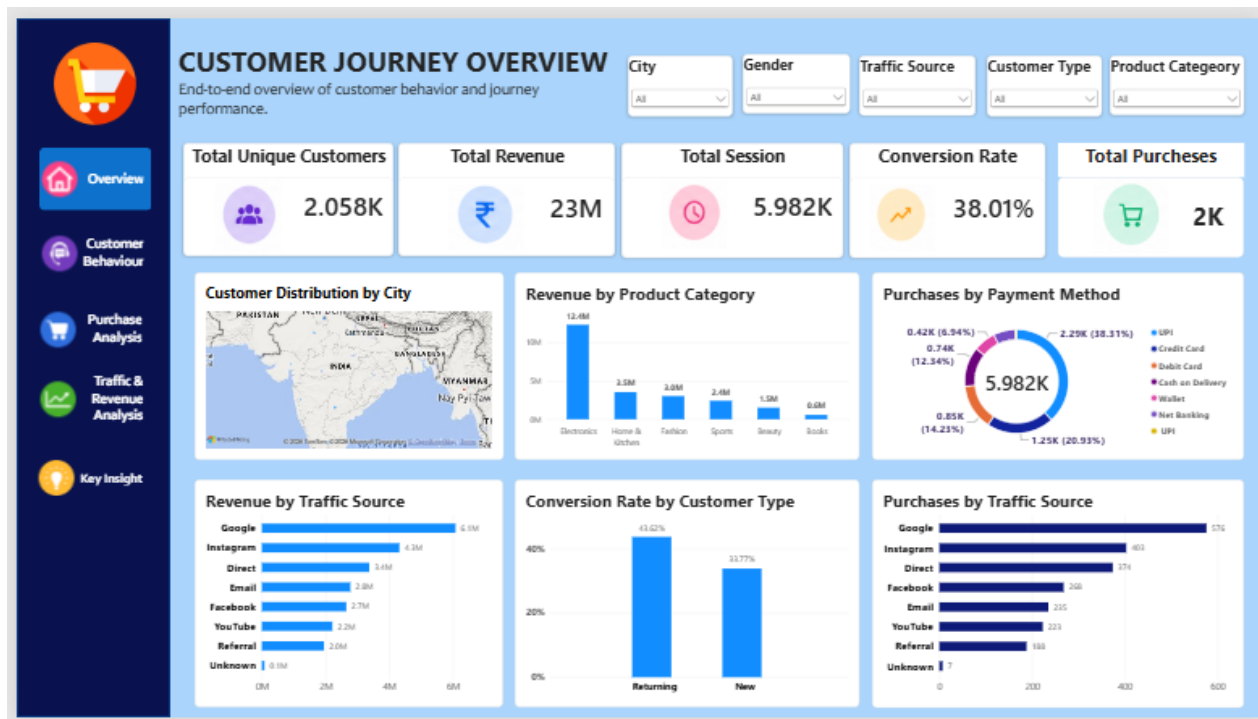


Figure 23. Power BI — Customer Journey Overview

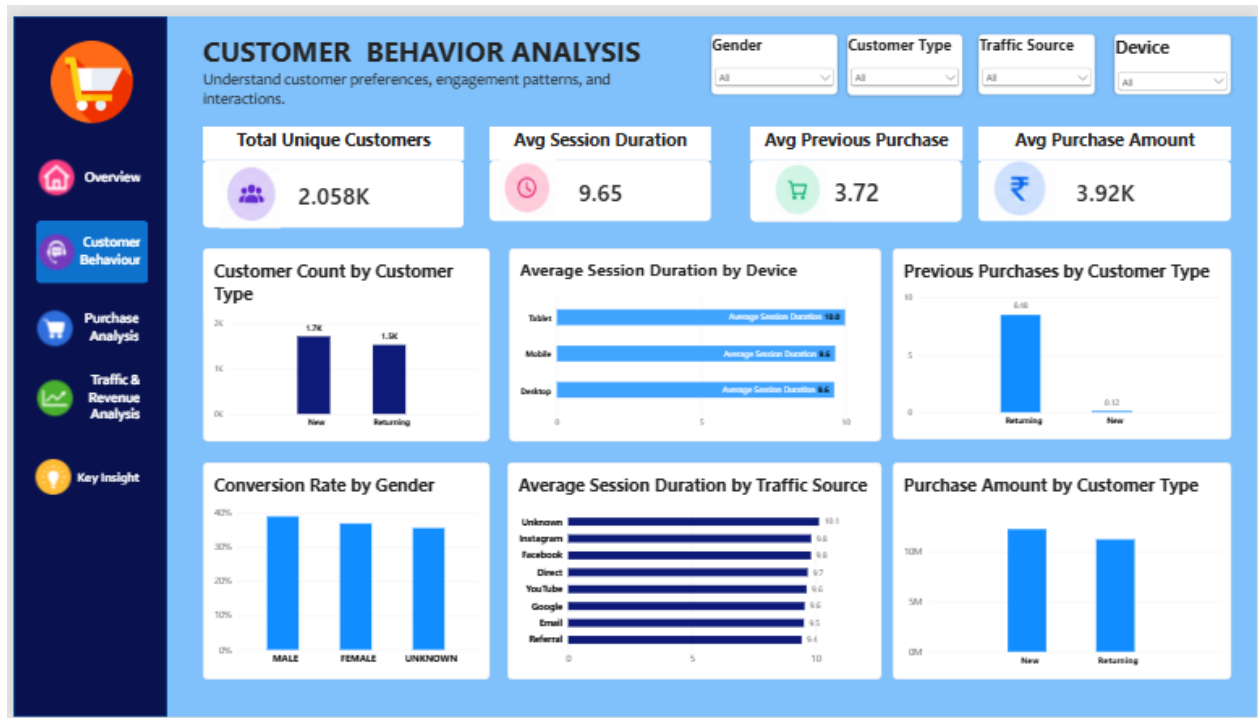


Figure 24. Power BI — Customer Behaviour Analysis

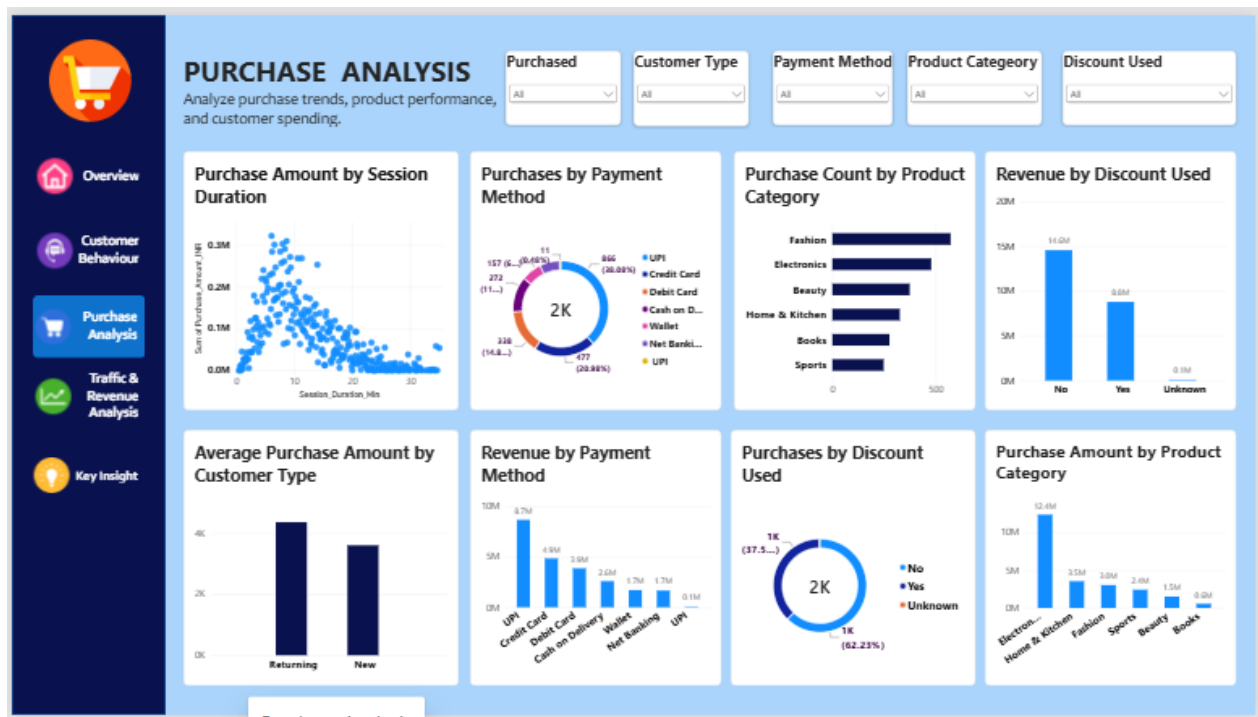


Figure 25. Power BI — Purchase Analysis

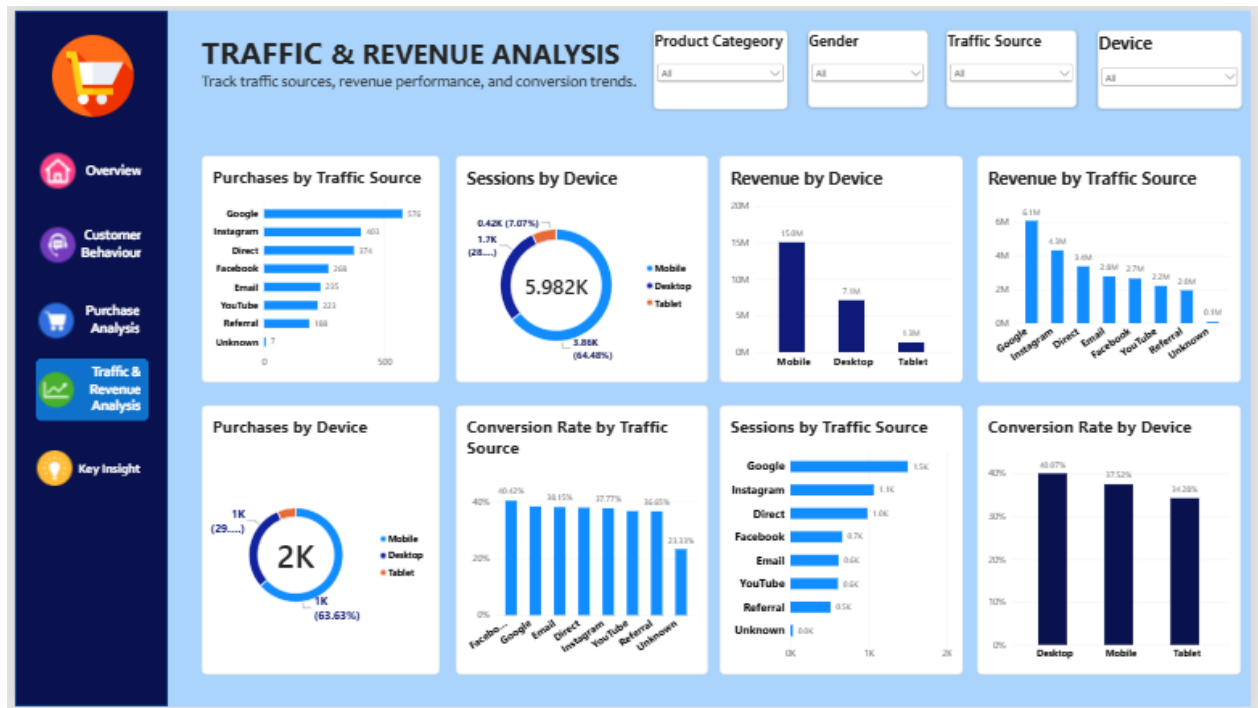


Figure 26. Power BI — Traffic & Revenue Analysis

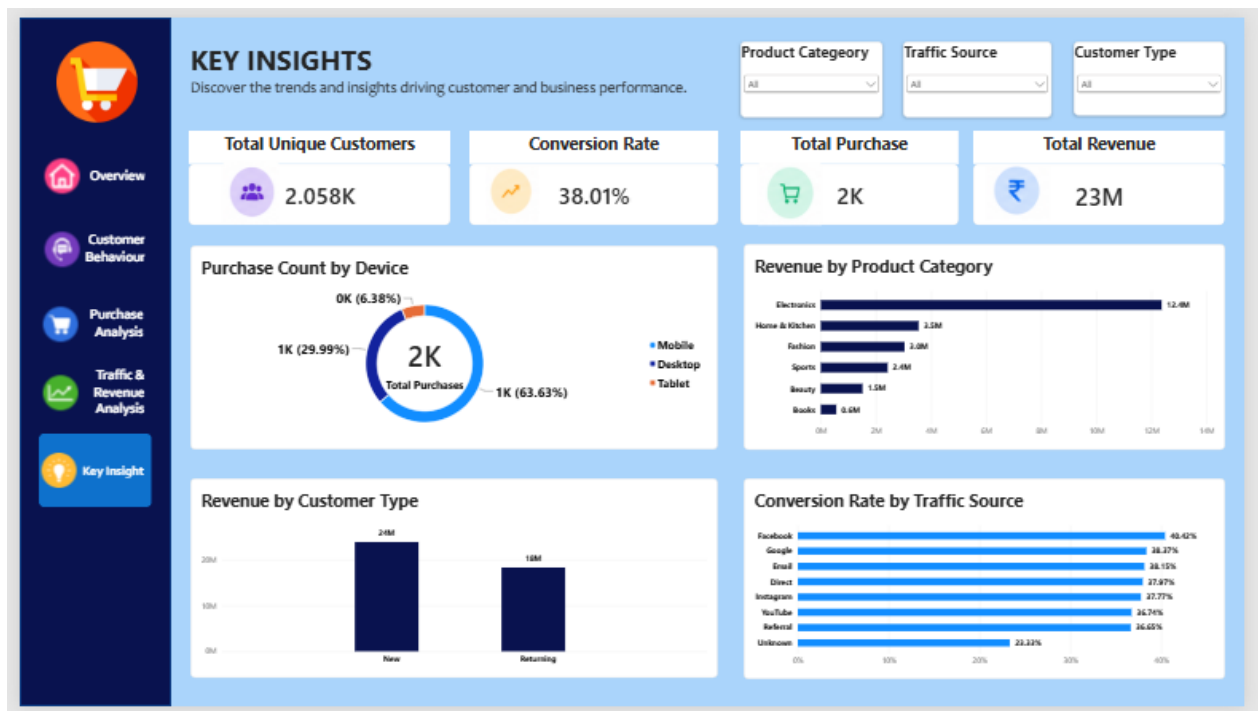


Figure 27. Power BI — Key Insights

12. Dashboard Page-by-Page Relevance

Page	Primary Purpose	Business Relevance
1. Customer Journey Overview	Executive view of customers, sessions, purchases, conversion and revenue.	Fast assessment of overall customer-journey health and commercial performance.
2. Customer Behaviour Analysis	Customer type, session duration, previous purchases, device, gender and traffic behaviour.	Segmentation and behavioural comparison between customer groups.
3. Purchase Analysis	Purchase amount, category purchases, payment methods, discount usage and revenue.	Understand transaction behaviour, product performance and value drivers.
4. Traffic & Revenue Analysis	Traffic-source and device sessions, purchases, conversion and revenue.	Evaluate acquisition-channel quality and device-level performance.
5. Key Insights	Concise summary of conversion, purchase, revenue, category, customer-type and traffic findings.	Management communication and decision support.

Page 1 — Customer Journey Overview

This is the executive starting point. It consolidates the major KPIs and broad comparisons across city, traffic source, customer type, product category and payment method.

Page 2 — Customer Behaviour Analysis

This page focuses on who the customers are and how they behave during sessions. It supports segmentation by customer type and comparisons across devices, gender and traffic sources.

Page 3 — Purchase Analysis

This page moves from behaviour into the transaction itself. It examines purchase amount, product-category volume, payment methods, discount usage and customer-type purchase value.

Page 4 — Traffic & Revenue Analysis

This page connects acquisition channels to commercial outcomes. Traffic volume can be evaluated against purchases, conversion and revenue, with device-level comparisons supporting journey optimization.

Page 5 — Key Insights

This is the decision-oriented summary layer. It brings the most important comparisons together for fast communication and final decision support.

13. Integrated Business Insights

13.1 Overall Conversion

The SQL analysis records 2,274 purchases across 5,982 sessions, resulting in an overall purchase conversion rate of approximately **38.01%**.

13.2 Returning Customers Convert Better

Returning customers show a **43.62%** conversion rate versus **33.77%** for new customers — a **9.85 percentage-point** gap.

13.3 Acquisition Performance Differs by Objective

Facebook has the highest displayed traffic-source conversion rate at **40.42%**, while Google generates the highest traffic-source revenue at **₹6,081,848**.

13.4 Electronics Dominates Revenue

Electronics contributes **₹12,366,407** of the **₹23,454,694** SQL revenue, or roughly **52.7%** of total revenue.

13.5 Fashion Leads Purchase Volume

Fashion records the highest category purchase count at **572**, while Electronics leads revenue.

13.6 Revenue Concentration

Revenue is concentrated in Electronics and high-contribution traffic sources such as Google and Instagram. This creates an opportunity for focused growth but also introduces concentration risk.

13.7 Device-Level Analysis Matters

The dashboard provides separate device comparisons for sessions, purchases, revenue and conversion across Mobile, Desktop and Tablet.

13.8 Data Quality Affects Interpretation

The payment-method SQL output contains two separate rows labelled UPI. This is a residual category-label inconsistency and should be standardized before formal decision-making.

14. Recommendations

1. Use returning-customer behaviour as a benchmark when evaluating retention and repeat-purchase initiatives.
2. Separate acquisition strategy by objective: evaluate Facebook strongly for conversion efficiency and Google strongly for revenue contribution.
3. Continue monitoring Electronics as the primary revenue category while assessing whether revenue concentration creates commercial risk.
4. Evaluate product-category purchase volume alongside revenue rather than using a single KPI.
5. Use traffic-source and device-level conversion together to identify opportunities for improving the customer journey.
6. Strengthen categorical standardization rules in future data pipelines, including payment-method labels.
7. Maintain explicit business rules for structurally non-applicable fields such as reviews and satisfaction responses.

8. Use the Power BI dashboard as a monitoring layer and refresh the analytical dataset periodically for trend comparison.

15. Data Quality Considerations

- The revised analytical dataset contains 5,982 records used for SQL and Power BI analysis.
- Product Category, Gender and City required consistency checks and standardization.
- Invalid Review_Rating values identified during cleaning were converted to missing values.
- Remaining Review_Rating and Customer_Satisfaction nulls are largely explained as structurally non-applicable to non-purchasing sessions.
- Payment-method reporting contains a visible duplicate UPI label, which should be normalized in a future cleaning iteration.
- The earlier 6,000-row intermediate validation view is not used as the final analytical record count in this revised report.

16. Limitations

- The project is descriptive and diagnostic; it does not establish causal relationships.
- No predictive modelling or machine-learning stage is evidenced in the supplied project material.
- Separate Power Query, DAX and Power BI data-model evidence is not available in the supplied project material.
- External market, campaign-cost or acquisition-cost data is not incorporated.
- Some dashboard KPIs are displayed in rounded/abbreviated form, while SQL outputs provide more precise values.
- The supplied evidence does not clearly provide final result screenshots for SQL Questions 1 and 2, so those outputs are not reconstructed.

17. Future Scope

- Introduce time-series trend analysis if reliable date history is available.
- Build funnel analysis from session engagement to product interaction, cart activity and purchase.
- Add customer retention and cohort analysis.
- Create automated data-quality rules and repeatable ETL/data-preparation workflows.
- Develop predictive purchase-propensity models after establishing a stable historical dataset.
- Add channel ROI or acquisition-cost data where available.
- Extend the dashboard with alerting and scheduled refresh for operational monitoring.

18. Conclusion

Customer Journey Analytics Dashboard demonstrates an end-to-end individual data-analytics workflow from raw customer journey data to SQL-driven business analysis and interactive Power BI reporting. The project establishes a clear sequence: understand the data, profile quality, clean and standardize important fields, define the analytical dataset, analyse business questions in MySQL, and communicate the findings through five focused dashboard pages.

For this revised submission, the analytical record count is explicitly **5,982**, because that is the dataset loaded into Power BI and used for the computations. The SQL purchase-status outputs reconcile to this same session population: 2,274 purchased sessions plus 3,708 non-purchased sessions.

The strongest findings are a 38.01% overall conversion rate, materially higher conversion among returning customers, Facebook leading traffic-source conversion, Google leading traffic-source revenue, Electronics leading category revenue and Fashion leading purchase volume.

Appendix A — Evidence Figures

This appendix provides the curated evidence trail from the supplied project material. Figures that visibly showed an earlier 6,000-row intermediate state have intentionally not been presented as proof of the final analytical record count.

Figure 1. Raw customer journey dataset

Figure 2. Dataset data dictionary

Figure 3. Initial dataset inspection

Figure 4. Dataset structure / data types

Figure 5. Descriptive statistics

Figure 6. Missing-value analysis

Figure 7. Product category before standardization

Figure 8. Product category after standardization

Figure 9. Gender before standardization

Figure 10. Gender after standardization

Figure 11. City before standardization

Figure 12. City after standardization

Figure 13. Duplicate inspection during cleaning

Figure 14. MySQL database setup

Figure 15. SQL — Purchased vs Not Purchased

Figure 16. SQL — Conversion Rate by Customer Type

Figure 17. SQL — Traffic Source Conversion

Figure 18. SQL — Product Category Performance

Figure 19. SQL — Revenue by Product Category

Figure 20. SQL — Revenue by Traffic Source

Figure 21. SQL — Successful Purchases by Payment Method

Figure 22. SQL — Overall Purchase and Revenue Performance

Figure 23. Power BI — Customer Journey Overview

Figure 24. Power BI — Customer Behaviour Analysis

Figure 25. Power BI — Purchase Analysis

Figure 26. Power BI — Traffic & Revenue Analysis

Figure 27. Power BI — Key Insights

Appendix B — SQL Evidence Summary

The supplied evidence clearly supports SQL analysis for Questions 3 through 10. Verified outputs are recorded below; missing Q1/Q2 outputs are not reconstructed.

SQL Area	Result
Q3 — Purchased vs Not Purchased	Yes: 2,274; No: 3,708
Q4 — Customer Type Conversion	New 33.77%; Returning 43.62%
Q5 — Traffic Source Conversion	Facebook 40.42% highest displayed conversion
Q6 — Product Category Performance	Fashion 572 purchases highest
Q7 — Revenue by Product Category	Electronics ₹12,366,407 highest
Q8 — Revenue by Traffic Source	Google ₹6,081,848 highest
Q9 — Successful Purchases by Payment Method	UPI-labelled rows; Credit Card 477; Debit Card 338; COD 272; Wallet 157; Net Banking 153
Q10 — Overall Performance	2,274 purchases; ₹23,454,694 revenue; ₹10,314.29 average purchase value

Appendix C — Project Traceability Checklist

Project Stage	Evidence / Status
Raw Excel	Documented
Dataset understanding	Documented
Data profiling	Documented
Missing-value analysis	Documented
Category standardization	Documented
Duplicate identification/removal	Documented
Final analytical dataset	5,982 records used for SQL and Power BI
MySQL import	Documented
SQL business analysis	Documented for available evidence
Power BI dashboard	Five pages documented
Business insights	Integrated from SQL and dashboard evidence
Undocumented Power BI internals	Intentionally not claimed

PROJECT AUTHOR

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Individual End-to-End Data Analytics Project
Customer Journey Analytics Dashboard
2026